REFLECTION JOURNAL

Original File Name

MT_Reflection_Journal_Alex_DeBusk.pdf

Reflection

Throughout this midterm project, I developed a better understanding of how data is prepared for machine learning and why each step of the preprocessing process is important. As someone who has only recently started learning Python, this project gave me an opportunity to become more familiar with how libraries such as Pandas, NumPy, and Matplotlib are used in data analysis. By reviewing the dataset and the preprocessing workflow, I gained a better understanding of how raw data can be transformed into a format suitable for machine learning.

One of the main concepts I learned was the importance of separating training and testing data before performing exploratory data analysis or preprocessing. Using a 70/30 split helps ensure that information from the testing data does not influence the training process, reducing the risk of data leakage. I also learned about different preprocessing techniques, including median imputation for missing values, ordinal and one-hot encoding for categorical variables, and standardization for numerical features. The use of a logarithmic transformation to address the highly skewed selling prices also helped me understand how data distributions can affect machine learning models.

Another important part of this project was learning more about Python's role in automating data preparation. Rather than manually changing values in a spreadsheet, Python allows data to be cleaned, transformed, and analyzed through code. I also gained a better understanding of how visualizations can be used to identify patterns, recognize unusual distributions, and compare data before and after preprocessing. These concepts helped me see how different Python functions and libraries work together as part of a larger data analysis process.

Overall, this project helped me become more familiar with the early stages of a machine learning workflow and the reasoning behind the techniques used to prepare a dataset. It also gave me a better idea of how regression models, such as Linear Regression, Ridge Regression, and Random Forest, can be used to predict used car prices. The knowledge gained from this project will be useful as I continue developing my Python skills and move toward training and evaluating machine learning models for the final project.

REFLECTION JOURNAL

Original File Name

MT_Reflection_Journal_Shareefah_Adeola_ITAI1371.pdf

Reflection Journal

Student: Shareefah Adeola

Team: ML_1371_14274 4_ITAI1371

Date: 10/03/2026

Reflection on the Used-Car Machine Learning Project

Working on the used-car dataset project has helped me better understand the importance of data preparation before applying machine-learning techniques. I learned that a dataset must be carefully examined and cleaned before it can be used for analysis or modeling.

One of my contributions to the team was helping organize the vehicle information by adding vehicle type as a manageable grouping variable. This process started when the original data was pulled and shared with the professor. The professor pointed out that having such many vehicle types/categories would be too much to work effectively for our analysis. As a result, we introduced vehicle type as the first column to help organize the vehicles into more manageable groups.

The initial dataset contained approximately 8,125 rows. We selected approximately 4,022 rows after introducing vehicle type as a manageable grouping variable. This helped us organize the data into more reasonable groups because the original dataset contained many different vehicle makes, which would have resulted in too many categories to work with effectively.

During the preprocessing stage, I learned how to identify and remove duplicate records, identify missing values, and replace missing numerical values using the median. I also learned how to identify extreme values using the IQR method and how capping extreme km_driven values can reduce their influence without removing the observations from the dataset.

Another important part of the project was learning how categorical variables can be converted into numerical variables using one-hot encoding. I also learned how to standardize numerical variables using StandardScaler so that the features are placed on a comparable scale.

The project also introduced me to feature engineering. We created a vehicle_age variable from the vehicle year using 2020 as the reference year. This helped me understand how an existing variable can be transformed into a new feature that may provide more useful information for machine-learning analysis.

One challenge I experienced was learning to work with Jupyter Notebook, which I had not used before. In my previous coursework, I had primarily used Google Colab. I also experienced challenges when working with Python and Google Colab, particularly when errors occurred or when the runtime restarted and the variables were lost. This taught me the importance of running notebook cells in the correct order and saving important, processed datasets. I also learned that preprocessing steps must be performed consistently between the training and testing datasets.

Overall, this project has improved my understanding of exploration data analysis, data cleaning, preprocessing, feature engineering, categorical encoding, and feature scaling. It has also helped me understand that the quality of the data preparation process is an important part of developing a reliable machine-learning project.

REFLECTION JOURNAL

Original File Name

MT_Reflection_Journal_Yilin_Leng_ITAI1371.pdf

Reflection

For this midterm project, I worked through the main steps of preparing the used car dataset for machine learning. I started by separating the data into 70% for training and 30% for testing before beginning EDA or handling missing values. Keeping the test set separate helps prevent information from the test data from affecting the preprocessing process.

During the data cleaning stage, I noticed that the selling price values were not evenly distributed, so I used a log transformation to reduce the effect of the highly skewed values. I also standardized the continuous numerical features and converted categorical information into numerical forms. The owner feature was handled with ordinal encoding, while variables such as fuel type and transmission were processed with one hot encoding. For missing values in numerical features such as mileage, engine, Maxpower, and seats, I used median imputation. I also extracted the main car brand information from the original vehicle data to make the feature set easier to work with.

Working on this project helped me understand that preprocessing is more than just cleaning a dataset. The way the data is split, transformed, and organized can directly affect the results of a machine learning model. The processed training

data from this project will be used for the regression models in the final exam, where I will compare the models and work toward reducing their prediction error.